Module 2: Metric Spaces and Function Convergence

Topology, completeness, compactness, continuity, and convergence of functions

1. Metric Spaces

A *metric* abstracts the notion of distance without requiring an order or vector-space structure. A *metric space* is a pair (X, d), where X is a set and $d : X \times X \to [0, \infty)$ satisfies, for every $x, y, z \in X$:

(i) **Definiteness:** $d(x, y) = 0 \iff x = y$ (and $d(x, y) \geq 0$).
(ii) **Symmetry:** $d(x, y) = d(y, x)$.
(iii) **Triangle inequality:** $d(x, z) \leq d(x, y) + d(y, z)$.

Examples include $d(x, y) = |x - y|$ on $\mathbb{R}$, Euclidean distance $\|x - y\|_2$ on $\mathbb{R}^m$, and the *discrete metric* $d(x, y) = \mathbf{1}_{\{x \neq y\}}$ on any set. The same underlying set can carry different metrics, hence different notions of closeness.

Key point. A metric is additional *structure* on a set. Convergence, continuity, and completeness are then defined using this structure.

2. Topological Foundations

2.1 Open balls, open and closed sets

For $x \in X$ and $r > 0$, the *open ball* is

$$B_r(x) = \{y \in X : d(x, y) < r\}.$$

A subset $U \subseteq X$ is *open* if $\forall x \in U \, \exists r > 0$ such that $B_r(x) \subseteq U$. A set $F \subseteq X$ is *closed* if $X \setminus F$ is open. A *neighborhood* of x is any set containing an open ball about x.

2.2 Topology and its three axioms

A *topology* on X is a collection $\mathcal{T} \subseteq \mathcal{P}(X)$ whose members are called open sets and which satisfies

$$\boxed{\begin{aligned} &\varnothing, X \in \mathcal{T}, \\ &\bigcup_{i \in I} U_i \in \mathcal{T} \quad \text{for every family } (U_i)_{i \in I} \subseteq \mathcal{T}, \\ &U_1 \cap \cdots \cap U_k \in \mathcal{T} \quad \text{for every finite family in } \mathcal{T}. \end{aligned}}$$

The pair $(X, \mathcal{T})$ is a *topological space*. Every metric d induces a topology $\mathcal{T}_d$ via its open balls, but topology records which sets are open rather than exact numerical distances. Not every topology comes from a metric.

2.3 Interior, closure, and limit points

For $A \subseteq X$,

$$\operatorname{int}(A) = \bigcup \{U \subseteq A : U \text{ open}\}, \qquad \overline{A} = \bigcap \{F \supseteq A : F \text{ closed}\}.$$

Thus $\operatorname{int}(A)$ is the *largest open subset* of A, while $\overline{A}$ is the *smallest closed superset*. In a metric space,

$$\boxed{\begin{aligned} x \in \operatorname{int}(A) &\iff \exists r > 0 : \ B_r(x) \subseteq A, \\ x \in \overline{A} &\iff \forall r > 0 : \ B_r(x) \cap A \neq \varnothing. \end{aligned}}$$

A *limit point* (accumulation point) of A is a point x such that every $B_r(x)$ contains some point of A *other than* x:

$$x \in A' \iff \forall r > 0 : \ (B_r(x) \setminus \{x\}) \cap A \neq \varnothing.$$

In metric spaces, $\overline{A} = A \cup A'$; a set is closed iff it contains all its limit points. Also,

$$x \in \overline{A} \iff \exists (a_n) \subseteq A : \ a_n \to x.$$

For example, $A = \{1/n : n \geq 1\} \subset \mathbb{R}$ has $A' = \{0\}$ and $\overline{A} = A \cup \{0\}$.

3. Convergence and Completeness

Convergence. In (X, d), a sequence (x_n) converges to $x \in X$ if

$$\boxed{\forall \varepsilon > 0 \ \exists N \ \forall n \geq N : \ d(x_n, x) < \varepsilon.}$$

The metric axioms imply uniqueness of limits: if $x_n \to x$ and $x_n \to y$, then $d(x, y) \leq d(x, x_n) + d(x_n, y) \to 0$, so $x = y$.

Cauchy sequences. A sequence is *Cauchy* if its sufficiently late terms are uniformly close to one another:

$$\boxed{\forall \varepsilon > 0 \ \exists N \ \forall m, n \geq N : \ d(x_m, x_n) < \varepsilon.}$$

Every convergent sequence is Cauchy, by the triangle inequality. The converse depends on the space.

Completeness. The metric space (X, d) is *complete* if

$$\boxed{(x_n) \text{ Cauchy in } X \implies \exists x \in X : \ x_n \to x.}$$

For instance, $\mathbb{R}^m$ is complete with Euclidean distance, while $\mathbb{Q}$ with $|\cdot|$ is not: rational approximations to $\sqrt{2}$ are Cauchy but have no limit in $\mathbb{Q}$. Every closed subspace of a complete metric space is complete, and every complete subspace is closed in the ambient metric space.

Key point. Completeness means no *missing Cauchy limits*; it does not imply that arbitrary sequences converge or even have convergent subsequences.

4. Compactness

4.1 Open covers and compact sets

An *open cover* of $K \subseteq X$ is a family of open sets $(U_i)_{i \in I}$ with $K \subseteq \bigcup_{i \in I} U_i$. The set K is *compact* if every open cover admits a *finite subcover*:

$$\boxed{K \subseteq \bigcup_{i \in I} U_i \implies \exists i_1, \ldots, i_k : \ K \subseteq \bigcup_{j=1}^{k} U_{i_j}.}$$

Equivalently, every family of closed subsets of K with the *finite intersection property* has nonempty total intersection. This expresses a global finiteness principle rather than a numerical bound on diameter.

4.2 Heine–Borel and sequential compactness

Heine–Borel theorem. For the usual Euclidean metric on $\mathbb{R}^m$,

$$\boxed{K \subseteq \mathbb{R}^m : \quad K \text{ compact} \iff K \text{ closed and bounded.}}$$

This equivalence is special to finite-dimensional Euclidean spaces; it is false for general metric spaces.

Sequential compactness. For a metric space, compactness is equivalent to

$$\boxed{\text{Every sequence } (x_n) \subseteq K \text{ has a subsequence } x_{n_k} \to x \in K.}$$

One proof direction illustrates the role of compactness. Given $(x_n) \subseteq K$, let

$$F_N = \overline{\{x_n : n \geq N\}}^{K}, \qquad F_1 \supseteq F_2 \supseteq \cdots .$$

These are nonempty closed subsets of compact K. Every finite intersection is nonempty; compactness gives $x \in \bigcap_N F_N$. Hence, for every N and $r > 0$, some $n \geq N$ satisfies $d(x_n, x) < r$. Recursively choose $n_k > n_{k-1}$ with $d(x_{n_k}, x) < 1/k$; then $x_{n_k} \to x$.

4.3 Compactness, completeness, and total boundedness

A metric space is *totally bounded* if, for every $\varepsilon > 0$, finitely many radius-ε balls cover it:

$$\forall \varepsilon > 0 \ \exists x_1, \ldots, x_k \in X : \quad X = \bigcup_{j=1}^{k} B_\varepsilon(x_j).$$

For metric spaces, the precise structural characterization is

$$\boxed{\text{Compact} \iff \text{Complete and totally bounded.}}$$

Compactness implies completeness because a Cauchy sequence with a convergent subsequence must itself converge. It also implies total boundedness by taking a finite subcover of the ε-ball cover. Conversely, total boundedness allows extraction of a Cauchy subsequence; completeness supplies its limit.

Counterexample to "closed and bounded implies compact." An infinite set with the discrete metric is complete and bounded ($d \le 1$), but not totally bounded: for $0 < \varepsilon < 1$, each ball is a singleton, so finitely many balls cannot cover it.

Key point. Completeness prevents missing limits; total boundedness prevents infinitely many mutually separated points at a fixed scale. Together they yield compactness.

5. Compactness and Continuity

5.1 Continuity and compact images

For $f : (X, d_X) \to (Y, d_Y)$, continuity at $x \in X$ means

$$\forall \varepsilon > 0 \ \exists \delta > 0 : \ d_X(x, y) < \delta \Rightarrow d_Y(f(x), f(y)) < \varepsilon.$$

Equivalently, *topological continuity* requires the inverse image of every open set in Y to be open in X:

$$\boxed{f \text{ continuous} \iff V \text{ open in } Y \Rightarrow f^{-1}(V) \text{ open in } X.}$$

The key global consequence is that *continuous images of compact sets are compact.* Indeed, pull an open cover of $f(K)$ back along f, take a finite subcover of K, then push the covering statement forward to $f(K)$.

5.2 Extreme Value Theorem

If $K \neq \varnothing$ is compact and $f : K \to \mathbb{R}$ is continuous, then $f(K)$ is compact in $\mathbb{R}$, hence closed and bounded. Consequently,

$$\boxed{\exists x_{\min}, x_{\max} \in K : \quad f(x_{\min}) = \min_{x \in K} f(x), \qquad f(x_{\max}) = \max_{x \in K} f(x).}$$

Compactness guarantees not only finite bounds but also *attainment* of both extrema.

5.3 Uniform continuity and Heine–Cantor

Ordinary continuity allows δ to depend on the base point x. *Uniform continuity* requires one δ to work across all points:

$$\boxed{\forall \varepsilon > 0 \ \exists \delta > 0 \ \forall x, y \in X : \ d_X(x, y) < \delta \Rightarrow d_Y(f(x), f(y)) < \varepsilon.}$$

Heine–Cantor theorem. Every continuous map from a compact metric space to a metric space is uniformly continuous. *Proof idea:* if uniform continuity fails, choose $x_n, y_n \in K$ with $d(x_n, y_n) < 1/n$ but $d_Y(f(x_n), f(y_n)) \ge \varepsilon_0$. Compactness gives $x_{n_k} \to x$; then $y_{n_k} \to x$, contradicting continuity at x.

6. Convergence of Function Sequences

6.1 Pointwise versus uniform convergence

Let $f_n : X \to \mathbb{R}$. *Pointwise convergence* to f means

$$\forall x \in X \ \forall \varepsilon > 0 \ \exists N = N(x, \varepsilon) \ \forall n \ge N : \ |f_n(x) - f(x)| < \varepsilon.$$

Uniform convergence requires N to be independent of x:

$$\boxed{\forall \varepsilon > 0 \ \exists N = N(\varepsilon) \ \forall n \ge N \ \forall x \in X : \ |f_n(x) - f(x)| < \varepsilon.}$$

Uniform convergence implies pointwise convergence, but not conversely. For $f_n(x) = x^n$ on $[0, 1]$, the pointwise limit is $f(x) = 0$ for $x < 1$ and $f(1) = 1$. Yet

$$\sup_{x \in [0,1]} |f_n(x) - f(x)| = 1 \quad \text{for every } n,$$

so convergence is not uniform; continuous functions can have a discontinuous pointwise limit.

6.2 Uniform convergence as metric convergence

Let $B(X)$ be the space of bounded real-valued functions on X, with the *supremum metric*

$$d_\infty(f,g) = \|f-g\|_\infty := \sup_{x\in X} |f(x)-g(x)|.$$

Then

$$\boxed{f_n \xrightarrow{\text{uniform}} f \iff d_\infty(f_n,f)\to 0.}$$

Thus uniform convergence is ordinary convergence in an appropriate *function metric space*. The space $(B(X), d_\infty)$ is complete; in particular, $C([a,b])$ is complete under $\|\cdot\|_\infty$ since uniform limits of continuous functions remain continuous.

6.3 Interchanging limits with continuity and integration

Uniform Limit Theorem. If every f_n is continuous and $f_n \to f$ uniformly, then f is continuous. The proof uses

$$|f(x)-f(x_0)| \le |f(x)-f_n(x)| + |f_n(x)-f_n(x_0)| + |f_n(x_0)-f(x_0)|.$$

Uniform convergence controls the first and third terms with a *single fixed* n, while continuity of f_n controls the middle term.

Interchanging limit and integral. For continuous $f_n : [a,b] \to \mathbb{R}$ with uniform limit f,

$$\boxed{\left|\int_a^b f_n - \int_a^b f\right| \le (b-a)\|f_n - f\|_\infty \to 0, \qquad \lim_n \int_a^b f_n = \int_a^b \lim_n f_n.}$$

Pointwise convergence alone is insufficient. The continuous spikes

$$f_n(x) = \begin{cases} 6n^2 x(1-nx), & 0 \le x \le 1/n, \\ 0, & 1/n < x \le 1, \end{cases}$$

satisfy $f_n(x) \to 0$ for every $x \in [0,1]$, while $\int_0^1 f_n(x)\,dx = 1$ for every n. Uniform convergence is *sufficient, not necessary*: $x^n \to \mathbf{1}_{\{1\}}$ pointwise on $[0,1]$ and its integrals still tend to 0.

Key point. The mode of convergence determines which operations survive taking limits. Later, measure-theoretic convergence theorems replace uniform convergence by other useful hypotheses.